

An Iwatsubo-Based Solution for Labyrinth Seals: Comparison to Experimental Results¹

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The basic equations are derived for compressible flow in a labyrinth seal. The flow is assumed to be completely turbulent in the circumferential direction where the friction factor is determined by the Blasius relation. Linearized zeroth and first-order perturbation equations are developed for small motion about a centered position by an expansion in the eccentricity ratio. The zeroth-order pressure distribution is found by satisfying the leakage equation while the circumferential velocity distribution is determined by satisfying the momentum equation. The first-order equations are solved by a separation of variable solution. Integration of the resultant pressure distribution along and around the seal defines the reaction force developed by the seal and the corresponding dynamic coefficients. The results of this analysis are compared to published test results.

Introduction

The problem of self-excited vibration in turbomachinery due to labyrinth seals has led to the development of many analyses which attempt to model the physical phenomenon so that the problem can be better understood and therefore solved. The shortcoming with the analyses which have been presented to date is that they are difficult to understand and require limiting assumptions such as ignoring the area derivative in the circumferential direction, assuming that the friction factor is the same for all surfaces, and assuming that the flow coefficient is constant along the seal. These assumptions may be of some use mathematically, but do very little for the understanding of the physical occurrence. The first steps toward analysis of this problem were taken by Alford [1], who neglected circumferential flow, and Spürk et al. [2], who neglected rotation of the shaft. Vance and Murphy [3] extended the Alford analysis by introducing a more realistic assumption of choked flow. Kostyuk [4] performed the first comprehensive analysis, but failed to include the change in area due to eccentricity which is responsible for the relationship between cross-coupled forces and parallel rotor displacements. Iwatsubo [5, 6] refined the Kostyuk model by including the time dependency of area change, but he neglected the area derivative in the circumferential direction. Kurohashi [7] incorporated dependency of the flow coefficient on eccentricity into his analysis, but assumed that the circumferential velocity in each cavity was the same.

The analysis presented here includes the variation of the area in the circumferential direction due to eccentricity and incorporates as many of the physical phenomena in the flow field as was thought necessary to produce an adequate result.

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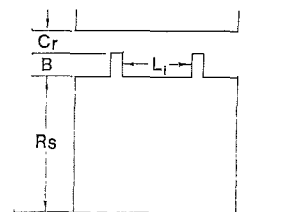


Fig. 1 A typical cavity

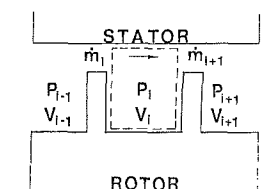


Fig. 2 Cavity control volume

The main purpose of this paper is to present a unified and comprehensive derivation of a reduced set of equations and a new solution format for those equations. The results of this analysis are compared with the published test results of Wachter and Benckert [8, 9, 10].

Procedure

The analysis presented here is developed for the see-through type of labyrinth seal shown in Fig. 1. The continuity and momentum equations will be derived for a single cavity control volume as shown in Figs. 2, 3, 4, and 5. A leakage model will be employed to account for the axial leakage. The governing equations will be linearized using perturbation analysis for small motion about a centered position. The zeroth-order continuity and momentum equations will be solved to determine the steady-state pressure and velocity for each cavity. The first-order continuity and momentum equations will be reduced to linearly independent algebraic

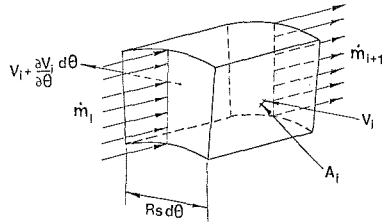


Fig. 3 Cavity control volume

equations by assuming an elliptical orbit for the shaft and a corresponding harmonic response for the pressure and velocity perturbations. The force and force coefficients for the seal are found by integration of the first-order pressure perturbation along and around the shaft.

Assumptions

- 1 Fluid is considered to be an ideal gas.
- 2 Pressure variations within a chamber are small compared to the pressure difference across a seal strip.
- 3 The frequency of acoustic resonance in the cavity is much higher than that of the rotor speed.
- 4 Added mass terms are neglected.
- 5 The eccentricity of the rotor is small compared to the radial seal clearance.
- 6 In the determination of the shear stresses in the circumferential direction, the axial component of velocity is neglected.
- 7 The contribution of shear stress to the stiffness and damping coefficients is neglected.
- 8 Constant temperature.

Governing Equations

Continuity Equation. Referring to the control volume in Figs. 2 and 3, the continuity equation for the control volume shown is:

$$\frac{\partial}{\partial t} (\rho_i A_i) + \frac{\partial}{\partial \theta} \left(\frac{\rho_i V_i A_i}{R_s} \right) + \dot{m}_{i+1} - \dot{m}_i = 0 \quad (1)$$

Nomenclature

A_i	= cross-sectional area of the cavity (L^2); defined in equation (1.a)
B_i	= height of labyrinth seal strip (L); defined in Fig. 1
C	= direct damping coefficient (Ft/L)
Cr	= nominal radial clearance (L); defined in Fig. 1
D_h	= hydraulic diameter of cavity (L); introduced in equation (3)
H	= local radial clearance (L)
K	= direct stiffness coefficient (F/L)
L	= pitch of seal strips (L); defined in Fig. 1
NT	= number of seal strips
$NC = NT - 1$	= number of cavities
P	= pressure (F/L^2)
R	= gas constant
R_s	= radius of seal (L); defined in Fig. 1
T	= temperature (T)
$R_s \omega$	= surface velocity of rotor (L/t)
V_i	= average velocity of flow in circumferential direction (L/t)
a, b	= radial seal displacement components due to elliptical whirl (L); defined in equation (13)
ar	= dimensionless length upon which shear stress acts on rotor
as	= dimensionless length upon which shear stress acts on stator

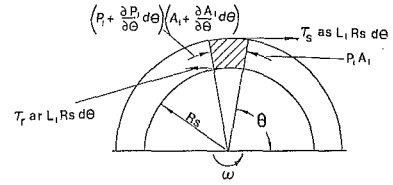


Fig. 4 Forces on control volume

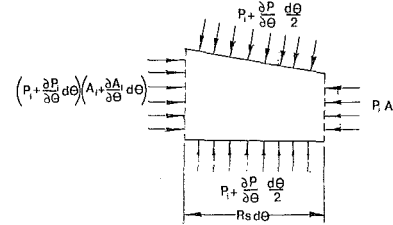


Fig. 5 Forces on control volume

where the transverse surface area A_i is defined by;

$$A_i = (B_i + H_i + B_{i+1} + H_{i+1})L_i/2 \quad (1.a)$$

Momentum Equation. The momentum equation (2) is derived using Figs. 4 and 5 which show the pressure forces and shear stresses acting on the control volume. This equation includes the area derivative in the circumferential direction, which was neglected by Iwatsubo [5, 6].

$$\frac{\partial \rho V_i A_i}{\partial t} + \frac{2\rho V_i A_i}{R_s} \frac{\partial V_i}{\partial \theta} + \frac{\rho V_i^2}{R_s} \frac{\partial A_i}{\partial \theta} + \frac{V_i A_i}{R_s} \frac{\partial \rho}{\partial \theta} + \dot{m}_{i+1} V_i - \dot{m}_i V_{i-1} = - \frac{A_i}{R_s} \frac{\partial P_i}{\partial \theta} + \tau_{ri} ar_i L_i - \tau_{si} as_i L_i \quad (2)$$

where ar and as are the dimensionless length upon which the shear stresses act and are defined for teeth on rotor by

$$as_i = 1 \quad ar_i = (2B_i + L_i)/L_i,$$

and for teeth on the stator by

c	= cross-coupled damping coefficient (Ft/L); in equation (18)
k	= cross-coupled stiffness coefficient (F/L); in equation (18)
$\dot{m}$	= leakage mass flow rate per circumferential length (M/Lt)
mr, nr, ms, ns	= coefficients for Blasius relation for friction factor; defined in equation (3)
t	= time (t)
ω	= shaft angular velocity ($1/t$)
ρ	= density of fluid (M/L^3)
ν	= kinematic viscosity (L^2/t)
ϵ	= e/Cr = eccentricity ratio
γ	= ratio of specific heats
π	= 3.141592
K_Q^*	= dimensionless cross-coupled stiffness parameter; defined in equations (27)
E_0^*	= dimensionless entry swirl parameter; defined in equations (27)

Subscripts

0	= zeroth-order component
i	= i -th chamber value
1	= first-order component
x	= X-direction
y	= Y-direction
r	= reservoir value
s	= sump value

$$as_i = (2B_i + L_i) / L_i \quad ar_i = 1$$

Blasius [11] determined that the shear stresses for turbulent flow in a smooth pipe could be written as

$$\tau = \frac{1}{2} \rho U_m^2 mo \left(\frac{Um Dh}{\nu} \right)^{mo}$$

where Um is the mean flow velocity relative to the surface upon which the shear stress is acting. The constants mo and no can be empirically determined for a given surface from pressure flow experiments. However, for smooth surfaces the coefficients given by Yamada [12] for turbulent flow between annular surfaces are:

$$mo = -0.25 \quad no = 0.079$$

Applying Blasius' equation to the labyrinth rotor and stator surfaces yields the following definitions for the rotor and stator shear stresses

$$\begin{aligned} \tau_{ri} &= \frac{\rho_i}{2} (R\omega - V_i)^2 nr \left[\frac{|R\omega - V_i| Dh_i}{\nu} \right]^{mr} \text{sgn}(R\omega - V_i) \\ \tau_{si} &= \rho_i \frac{V_i^2}{2} ns \left[\frac{|V_i| Dh_i}{\nu} \right]^{ms} \text{sgn}(V_i) \end{aligned} \quad (3)$$

where Dh_i is the hydraulic diameter defined by

$$Dh_i = \frac{2(H_i + B_i)L_i}{(H_i + B_i + L_i)} \quad (4)$$

Separate parameters (ms , ns), (mr , nr) for the stator and rotor, respectively, account for different rotor and stator roughnesses.

If equation (1) times the circumferential velocity is now subtracted from equation (2), the following reduced form of the momentum equation is obtained:

$$\begin{aligned} \rho_i A_i \frac{\partial V_i}{\partial t} + \rho_i V_i A_i \frac{\partial V_i}{R s \partial \theta} + \dot{m}_i (V_i - V_{i-1}) \\ = -\frac{A_i}{R s} \frac{\partial P_i}{\partial \theta} - \tau_{si} as_i L_i + \tau_{ri} ar_i L_i \end{aligned} \quad (5)$$

In order to reduce the numbers of variables, all of the density terms are replaced with pressure terms using the ideal gas law.

$$P_i = \rho_i R T \quad (6)$$

Furthermore, in order to make the perturbation analysis easier, the following substitution is made in the continuity equation:

$$\dot{m}_{i+1} - \dot{m}_i = \frac{\dot{m}_i^2 - \dot{m}_0^2}{2\dot{m}_0}$$

where $\dot{m}_0$ is the steady-state mass flow rate.

Leakage Equation. To account for the leakage mass flow rate in the continuity and momentum equations, the leakage model of Neumann [13] was chosen. This model predicts leakage and pressures fairly accurately and has a term to account for kinetic-energy carryover. However, the empirical flow coefficient relations given by Neumann were discarded in favor of the equations of Chaplygin [14] for flow through an orifice. This was done to produce a different flow coefficient for succeeding contractions along the seal as has been shown to be the case by Egli [15]. The form of the model is:

$$\dot{m}_i = \mu_{1i} \mu_2 H_i \sqrt{\frac{P_{i-1}^2 - P_i^2}{RT}} \quad (7)$$

where the kinetic-energy carryover coefficient μ_2 is defined for straight through seals as:

$$\mu_2 = \sqrt{\frac{NT}{(1-j)NT+j}}$$

where

$$j = 1 - (1 + 16.6Cr/L)^{-2}$$

and is unity, by definition, for interlocking and combination groove seals. The flow coefficient is defined as:

$$\mu_{1i} = \frac{\pi}{\pi + 2 - 5s_i + 2s_i^2} \quad \text{where } s_i = \left(\frac{P_{i-1}}{P_i} \right)^{\frac{\gamma-1}{\gamma}} - 1$$

For choked flow, Fliegner's formula [16] will be used for the last seal strip. It is of the form:

$$\dot{m}_{NC} = \frac{0.510 \mu_2}{\sqrt{RT}} P_{NC} H_{NT} \quad (8)$$

Perturbation Analysis

For cavity i , the continuity equation (1), momentum equation (5), and leakage equation (7) are the governing equations for the variables V_i , P_i , $\dot{m}_i$. A perturbation analysis of these equations is to be developed with the eccentricity ratio, $\epsilon = e/Cr$, selected to be the perturbation parameter. The governing equations are expanded in the perturbation variables

$$\begin{aligned} P_i &= P_{0i} + \epsilon P_{1i} & H_i &= Cr_i + \epsilon H_1 \\ V_i &= V_{0i} + \epsilon V_{1i} & A_i &= A_0 + \epsilon KH_i \end{aligned}$$

where $\epsilon = e/Cr$ is the eccentricity ratio. The zeroth-order equations define the leakage mass flow rate and the circumferential velocity distribution for a centered position. The first-order equations define the perturbations in pressure and circumferential velocity due to a radial position perturbation of the rotor. Strictly speaking, the results are only valid for small motion about a centered position.

Zeroth-Order Solution. The zeroth-order leakage equation is

$$\dot{m}_{i+1} = \dot{m}_i = \dot{m}_0 \quad (9)$$

and is used to determine both the leakage rate $\dot{m}_0$ and pressure distribution for a centered position. The leakage rate is determined using either equation (7) or equation (8), depending on the operating conditions. To determine whether the flow is choked or not, assume that the pressure in the last cavity is equal to the critical pressure for choking. Using this pressure, find the leakage from equation (8) and then use equation (7) to determine the reservoir pressure necessary to produce this condition. Based on this pressure, a determination can be made whether the flow is choked or not. The associated pressure distribution is determined by employing the calculated leakage, along with a known boundary pressure, and solving equation (7) sequentially for each cavity pressure P_{0i} .

The zeroth-order circumferential-momentum equation is

$$\begin{aligned} \dot{m}_0 (V_{0i} - V_{0i-1}) &= (\tau_{ri0} ar_i - \tau_{si0} as_i) L_i; \\ i &= 1, 2, \dots, NC, \end{aligned} \quad (10)$$

From calculated pressures, the densities can be calculated at each cavity from equation (6), and the only unknowns remaining in equation (10) are the circumferential velocities V_{0i} . Given an inlet tangential velocity, a Newton-root-finding approach can be used to solve equation (10) for the i -th velocity, one cavity at a time. This is done starting at the first cavity and working downstream.

First-Order Solution. The governing first-order equations (11, 12) define the pressure and velocity fluctuations resulting from the seal clearance function. The continuity equation (11) and momentum equation (12) follow:

$$\begin{aligned} G_{1i} \frac{\partial P_{1i}}{\partial t} + G_{1i} \frac{V_{0i}}{R s} \frac{\partial P_{1i}}{\partial \theta} + G_{1i} \frac{P_{0i}}{R s} \frac{\partial V_{1i}}{\partial \theta} + G_{3i} P_{1i} \\ + G_{4i} P_{1i-1} + G_{5i} P_{1i+1} = -G_{6i} H_i - G_{2i} \frac{\partial H_i}{\partial t} \end{aligned}$$

$$-G_{2i} \frac{V_{0i}}{Rs} \frac{\partial H_i}{\partial \theta} \quad (11)$$

$$X_{li} \frac{\partial V_{li}}{\partial t} + \frac{X_{li} V_{0i}}{Rs} \frac{\partial V_{li}}{\partial \theta} + \frac{A_{0i}}{Rs} \frac{\partial P_{li}}{\partial \theta} + X_{2i} V_{li} - \dot{m}_0$$

$$V_{li-1} + X_{3i} P_{li} + X_{4i} P_{li-1} = X_{5i} H_i \quad (12)$$

where the X_i 's and G_i 's are defined in Appendix A. If the shaft center moves in an elliptical orbit, then the seal clearance function can be defined as:

$$\begin{aligned} \epsilon H_i &= -a \cos \omega t \cos \theta - b \sin \omega t \sin \theta \\ &= -\frac{a}{2} [\cos(\theta - \omega t) + \cos(\theta + \omega t)] - \frac{b}{2} [\cos(\theta - \omega t) \\ &\quad - \cos(\theta + \omega t)] \end{aligned} \quad (13)$$

The pressure and velocity fluctuations can now be stated in the associated solution format:

$$P_{li} = P_{ci}^+ \cos(\theta + \omega t) + P_{si}^+ \sin(\theta + \omega t) + P_{ci}^- \cos(\theta - \omega t) + P_{si}^- \sin(\theta - \omega t) \quad (14)$$

$$V_{li} = V_{ci}^+ \cos(\theta + \omega t) + V_{si}^+ \sin(\theta + \omega t) + V_{ci}^- \cos(\theta - \omega t) + V_{si}^- \sin(\theta - \omega t) \quad (15)$$

Substituting equations (13), (14), and (15) into equations (11) and (12) and grouping like terms of sines and cosines (as shown in Appendix B) eliminates the time and theta dependency and yields eight linear algebraic equations per cavity. The resulting system of equations of the i -th cavity is of the form:

$$\begin{aligned} [A_{i-1}](X_{i-1}) + [A_i](X_i) + [A_i + 1](X_{i+1}) \\ = \frac{a}{\epsilon} (B_i) + \frac{b}{\epsilon} (C_i) \end{aligned} \quad (16)$$

where

$$\begin{aligned} (X_{i-1}) &= (P_{si-1}^+, P_{ci-1}^+, P_{si-1}^-, P_{ci-1}^-, V_{si-1}^+, \\ &\quad V_{ci-1}^+, V_{si-1}^-, V_{ci-1}^-)^T \\ (X_i) &= (P_{si}^+, P_{ci}^+, P_{si}^-, P_{ci}^-, V_{si}^+, V_{ci}^+, V_{si}^-, V_{ci}^-)^T \\ (X_{i+1}) &= (P_{si+1}^+, P_{ci+1}^+, P_{si+1}^-, P_{ci+1}^-, V_{si+1}^+, \\ &\quad V_{ci+1}^+, V_{si+1}^-, V_{ci+1}^-)^T \end{aligned}$$

The A matrices and column vectors B and C are given in Appendix B. To use equation (16) for the entire seal solution, a system matrix must be formed which is block tridiagonal in the A matrices. The size of this resultant matrix is $(8NC \times 8NC)$ since pressure and velocity perturbations at the inlet and the exit are assumed to be zero. This system is easily solved by various linear equation algorithms, and yields a solution of the form:

$$\begin{aligned} P_{si}^+ &= \frac{a}{\epsilon} F_{asi}^+ + \frac{b}{\epsilon} F_{bsi}^+ \\ P_{si}^- &= \frac{a}{\epsilon} F_{asi}^- + \frac{b}{\epsilon} F_{bsi}^- \\ P_{ci}^+ &= \frac{a}{\epsilon} F_{aci}^+ + \frac{b}{\epsilon} F_{bci}^+ \\ P_{ci}^- &= \frac{a}{\epsilon} F_{aci}^- + \frac{b}{\epsilon} F_{bci}^- \end{aligned} \quad (17)$$

Determination of Dynamic Coefficient

The force-motion equations for a labyrinth seal are assumed to be of the form:

$$-\begin{Bmatrix} F_x \\ F_y \end{Bmatrix} = \begin{bmatrix} K & k \\ -k & K \end{bmatrix} \begin{Bmatrix} X \\ Y \end{Bmatrix} + \begin{bmatrix} C & c \\ -c & C \end{bmatrix} \begin{Bmatrix} \dot{X} \\ \dot{Y} \end{Bmatrix} \quad (18)$$

The solution of equation (18) for the stiffness and damping

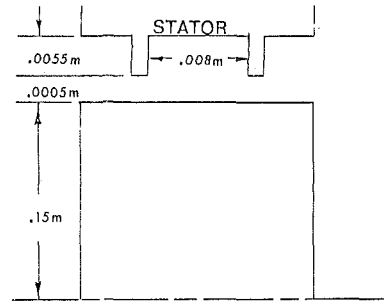


Fig. 6 Configuration used for experiment

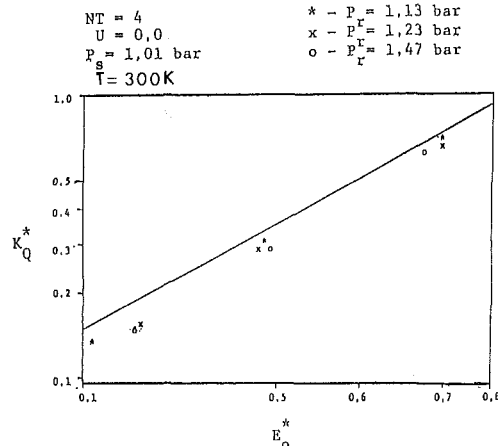


Fig. 7 Comparison to data of [8] for seal in Fig. 6

coefficients is the objective of the current analysis. For the assumed elliptical orbit of equation (13), the X and Y components of displacement and velocity are defined as:

$$\begin{aligned} X &= a \cos \omega t & \dot{X} &= -a \omega \sin \omega t \\ Y &= b \sin \omega t & \dot{Y} &= b \omega \cos \omega t \end{aligned}$$

Substituting these relations into (18) yields:

$$\begin{aligned} +F_x &= -Ka \cos \omega t - kb \sin \omega t + Ca \omega \sin \omega t - cb \omega \cos \omega t \\ +F_y &= -ka \cos \omega t - Kb \sin \omega t - ca \omega \sin \omega t - Cb \omega \cos \omega t \end{aligned} \quad (19)$$

Redefining the forces, F_x and F_y , as the following:

$$\begin{aligned} F_x &= F_{xc} \cos \omega t + F_{xs} \sin \omega t \\ F_y &= F_{yc} \cos \omega t + F_{ys} \sin \omega t \end{aligned} \quad (20)$$

and substituting back into (19) yields the following relations:

$$\begin{aligned} -F_{xc} &= Ka + cb \omega & -F_{xs} &= -Ca \omega + kb \\ -F_{yc} &= ka + Cb \omega & -F_{ys} &= Kb + ca \omega \end{aligned} \quad (21)$$

The X and Y components of force can be found by integrating the pressure around the seal as follows:

$$F_x = -Rs \epsilon \sum_{i=1}^{NC} \int_0^{2\pi} P_{li} L_i \cos \theta d\theta \quad (22)$$

$$F_y = Rs \epsilon \sum_{i=1}^{NC} \int_0^{2\pi} P_{li} L_i \sin \theta d\theta \quad (23)$$

Only one of these components needs to be expanded in order to determine the dynamic coefficients. For this analysis, the X component was chosen. Substituting equation (14) into (22) and integrating yields:

$$F_x = -\epsilon \pi Rs \sum_{i=1}^{NC} L_i [(P_{si}^+ - P_{si}^-) \sin \omega t + (P_{ci}^+ + P_{ci}^-) \cos \omega t] \quad (24)$$

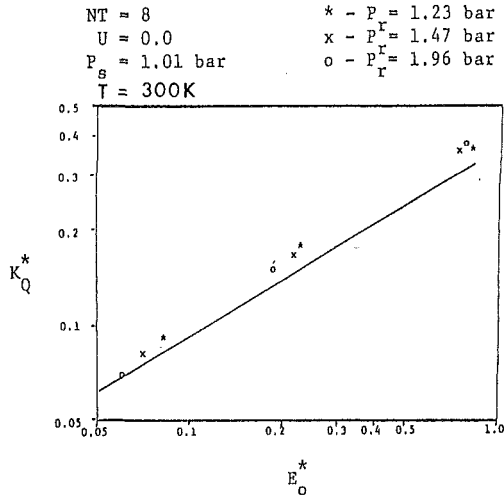


Fig. 8 Comparison to data of [8] for seal in Fig. 6

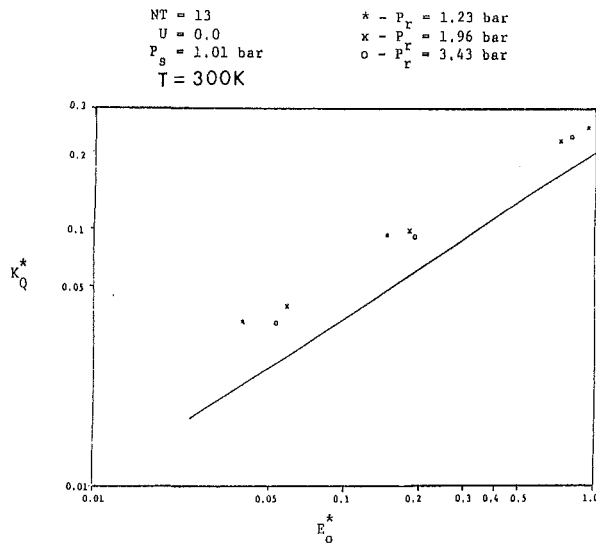


Fig. 9 Comparison to data of [8] for seal in Fig. 6

Substituting from equation (17) and (19) into equation (24) and equating coefficients of $\sin\omega t$ and $\cos\omega t$ yields:

$$F_{xs} = -\pi R s \sum_{i=1}^{NC} L_i [a(F_{asi}^+ - F_{asi}^-) + b(F_{bsi}^+ - F_{bsi}^-)]$$

$$F_{xc} = -\pi R s \sum_{i=1}^{NC} L_i [a(F_{aci}^+ + F_{aci}^-) + b(F_{bci}^+ + F_{bci}^-)] \quad (25)$$

Equating the definitions for F_{xs} and F_{xc} provided by equations (21) and (25) and grouping like terms of the linearly independent coefficients a and b yields the final solutions to the stiffness and damping coefficients:

$$K = \pi R \sum_{i=1}^{NC} (F_{aci}^+ + F_{aci}^-) L_i$$

$$k = \pi R \sum_{i=1}^{NC} (F_{bsi}^+ - F_{bsi}^-) L_i$$

$$C = \frac{-\pi R s}{\omega} \sum_{i=1}^{NC} (F_{asi}^+ - F_{asi}^-) L_i$$

$$c = \frac{\pi R s}{\omega} \sum_{i=1}^{NC} (F_{bci}^+ + F_{bci}^-) L_i \quad (26)$$

Solution Procedure Summary

In review, the solution procedure uses the following sequential steps:

- Leakage is determined from equation (7) or (8).
- Pressure distribution is found using equation (7).
- Velocity distribution is determined using equation (10).
- A system equation is formed and solved using the cavity equation (16).
- Results of this solution, as defined in equations (17), are inserted into equation (26).

Results

To compare the present analytic solution with the experimental results of Wachter and Benckert [8, 9, 10], the following dimensionless parameters are introduced. The dimensionless cross-coupled stiffness and entry-swirl parameters are defined by Wachter and Benckert as:

$$K_Q^* = \frac{Cr K_{xy}}{RsLNC(Pr - Ps)} \quad E_0^* = \frac{0.5\rho_0 V_0^2}{(Pr - Ps) + 0.5\rho_0 V_x^2} \quad (27)$$

All of the results presented for comparison in this paper are for a seal with teeth on the stator, with entry swirl, and no shaft rotation. Although Wachter and Benckert published results for shaft rotation, the data for the operating conditions and seal geometry were insufficient for use in this study. The results in Figs. 7, 8, and 9 are from [8] and show the relationship between cross-coupled stiffness and the entry swirl, for a seal with strips on the stator and the geometry shown in Fig. 6. The line shown is the experimental result and the symbols are the results from this analytical model. These figures show that the model compares favorably to the experimental results in magnitude and the overall trend for various operating conditions. The figures also show that the model does not yield a consistently high or low result. Instead, the model tends to overpredict the value of the stiffness for a large number of strips and underpredict stiffness for a small number of seals. This trend is probably due to errors in calculating the zeroth-order pressure distribution using the leakage model.

The results in Table 1 are from [9, 10] for a seal with strips on the stator. The results show the effect of change in seal parameters such as pitch, number of teeth, radius, strip height, and clearance on the cross-coupled stiffness. The model accurately shows the increase in cross-coupled stiffness due to decrease in clearance and decrease in strip height, but it fails to remain constant for change of pitch and consistently overestimates the cross-coupled stiffness for the larger radius cases by about 26%. The data in Table 2 are also from [9, 10] and show the results for full labyrinths. The comparison is very poor.

Conclusion

A clear and understandable analysis utilizing reduced equations has been presented for the problem of calculating rotordynamic coefficients for labyrinth seals. This paper was developed to provide a less restrictive analysis and a better explanation of the current analyses. The model developed gives results that are within 25% of the experimental results which are available. However, this error must be balanced against the known uncertainties in the experimental data. This is especially important since all of the data used are for a nonrotating shaft and the only influence on the cross-coupled stiffness was the entry swirl. Although Wachter and Benckert published data for a rotating shaft, the data were not sufficient to calculate a result. Also, the only data available for see-through labyrinths is for the type with strips on the stator. For a more rigorous test of this and other models, more complete data are required over a wider range of parameters

Table 1 Comparison of data for various geometries operating conditions for a seal with teeth on the stator and no shaft rotation [9, 10]

E_0^*	U	NT	$C_r(m)$	$L(m)$	$B(m)$	$R_s(m)$	$P_r(bar)$	$P_s(bar)$	EXP. $K_{xy}(N/mm)$	CALC. $K_{xy}(N/mm)$	% Error
0.023	0.0	18	.00025	.005	.0025	0.15	2.947	0.943	257	233	+9
0.024	0.0	18	.00025	.005	.0025	0.15	1.43	0.943	75	69	+8
0.038	0.0	18	.00025	.005	.006	0.15	2.947	0.943	157	142	+10
0.018	0.0	18	.00025	.005	.006	0.15	1.43	0.943	27	26	-4
0.04	0.0	18	.00058	.005	.006	0.075	1.925	0.943	29	20	-31
0.04	0.0	18	.00058	.005	.006	0.075	2.418	0.943	41	28	-32
0.04	0.0	9	.00058	.010	.006	0.075	1.925	0.943	29	16	-45
0.04	0.0	9	.00058	.010	.006	0.075	2.418	0.943	41	23	-44

Table 2 Comparison of data for full labyrinth [9, 10]

R_s mm	B mm	L mm	C_r mm	NT	W rpm	V_0 m/s	P_r/P_s	k(meas.) N/n	k(calc.) N/m	%Err
150	5.50	4.00	.500	18	9550	43.2	1.51	1.97 E5	0.19 E5	942
150	5.50	4.00	.500	24	9550	40.0	1.51	2.89 E5	0.38 E5	660
150	5.50	4.00	.500	24	9485	66.1	0.79	2.63 E5	0.31 E5	756
150	5.50	4.00	.500	24	9550	47.1	1.51	2.95 E5	0.44 E5	570

for different seal geometries. Finally, this analysis is only considered valid for the see-through type of labyrinth seal since the model fared very poorly in comparison with experimental results for interlocking seal data.

References

- 1 Alford, J. S., "Protecting Turbomachinery From Self-Excited Rotor Whirl," ASME JOURNAL OF ENGINEERING FOR POWER, Vol. 87, No. 4, Oct. 1965, pp. 333-344.
- 2 Spürk, J. H., and Keiper, R., "Selbsterregte Schwingungen bei Turbomaschinen infolge der Labyrinthströmung," *Ingenieur-Archiv* 43, 1974, pp. 127-135.
- 3 Vance, J. M., and Murphy, B. T., "Labyrinth Seal Effects on Rotor Whirl Stability," *Inst. of Mechanical Engineer*, 1980, pp. 369-373.
- 4 Kostyuk, A. G., "A Theoretical Analysis of the Aerodynamic Forces in the Labyrinth Glands of Turbomachines," *Teploenergetika*, 19 (11)0, 1972, pp. 39-44.
- 5 Iwatsubo, T., "Evaluation of Instability Forces of Labyrinth Seals in Turbines or Compressors," NASA CP 2133 Proceedings of a workshop at Texas A&M University 12-14 May 1980, entitled Rotordynamic Instability Problems in High Performance Turbomachinery, pp. 139-167.
- 6 Iwatsubo, T., Matooka, N., and Kawai, R., "Flow Induced Force and Flow Pattern of Labyrinth Seal," NASA CP 2250 Proceedings of a workshop at Texas A&M University 10-12 May 1982, entitled Rotordynamic Instability Problems in High Performance Turbomachinery, pp. 205-222.
- 7 Kurohashi, M., Inoue, Y., Abe, T., and Fujikawa, T., "Spring and Damping Coefficients of the Labyrinth Seal," Paper No. C283/80 delivered at the Second International Conference on Vibrations in Rotating Machinery, The Inst. of Mech. Engineering.
- 8 Wachter, J., and Benckert, H., "Querkraft aus Spaltdichtungen-Eine mögliche Ursache für die Laufunruhe von Turbomaschinen," *Atomkernenergie*, Vol. 32, 1978, Lfg. 4, pp. 239-246.
- 9 Wachter, J., and Benckert, H., "Flow Induced Spring Coefficients of Labyrinth Seals for Applications in Rotordynamic," NACA CP 2133 Proceedings of a workshop held at Texas A&M University 12-14 May 1980, entitled Rotordynamic Instability Problems of High Performance Turbomachinery, pp. 189-212.
- 10 Benckert, H., "Stromungsbedingte Federkennwerte in Labyrinthdichtungen," Doctoral dissertation, University of Stuttgart, 1980.
- 11 Blasius, H., "Forschungsbearb," *Ing.-Wes.*, No. 131, 1913.
- 12 Yamada, Y., *Trans. Japan Soc. Mechanical Engineers*, Vol. 27, No. 180, 1961, p. 1267.
- 13 Neumann, K., "Zur Frage der Verwendung von Durchblickdichtungen im Dampfturbinenbau," *Maschinentechnik*, Vol. 13, 1964, No. 4.
- 14 Gurevich, M. I., *The Theory of Jets in An Ideal Fluid*, Pergamon Press, 1966, pp. 319-323.
- 15 Egli, A., "The Leakage of Steam Through Labyrinth Glands," *Trans. ASME*, Vol. 57, 1935, pp. 115-122.
- 16 John, E. A. J., *Gas Dynamics*, Wiley, 1979.

APPENDIX A

Definition of the First-Order Continuity and Momentum Equation Coefficients

$$G_1 = \frac{A_{0i}}{RT} \quad G_2 = \frac{P_{0i}L}{RT}$$

$$G_3 = \frac{\dot{m}_0 P_i}{P_i^2 - P_{i-1}^2} + \frac{\dot{m}_0 \mu_{li+1}}{\pi} (5 - 4S_{li+1}) \frac{\gamma - 1}{\gamma P_{0i+1}}$$

$$\left(\frac{P_{0i}}{P_{0i+1}} \right)^{\frac{1}{\gamma}} + \frac{\dot{m}_0 \mu_{li}}{\pi} (5 - 4S_{li}) \left(\frac{\gamma - 1}{\gamma P_{0i}} \right) \left(\frac{P_{0i}}{P_{0i+1}} \right)^{\frac{\gamma - 1}{\gamma}}$$

$$G_4 = \frac{-\dot{m}_0 P_{0i-1}}{P_{0i-1}^2 - P_{0i}^2} + \frac{\dot{m}_0}{\pi} \mu_{li} (4S_i - 5)$$

$$\left[\frac{1}{P_{0i}} \left(\frac{\gamma - 1}{\gamma} \right) \left(\frac{P_{0i-1}}{P_{0i}} \right)^{-\frac{1}{\gamma}} \right]$$

$$G_5 = \frac{-\dot{m}_0 P_{0i+1}}{P_{0i}^2 - P_{0i+1}^2} + \frac{\dot{m}_0}{\pi} \mu_{li+1} (4S_i - 5)$$

$$\left[\frac{1}{P_{0i+1}} \left(\frac{\gamma - 1}{\gamma} \right) \left(\frac{P_{0i}}{P_{0i+1}} \right)^{\frac{\gamma - 1}{\gamma}} \right]$$

$$G_6 = \dot{m}_0 \left(\frac{C_{ri} - C_{ri+1}}{C_{ri+1} C_{ri}} \right)$$

$$X_1 = \frac{P_{0i} A_{0i}}{RT} \quad X_2 = \dot{m}_0 + \frac{\tau_{si} a s_i L_i (2 + ms)}{V_{0i}}$$

$$+ \frac{\tau_{ri} a r_i L_i (2 + mr)}{(R\omega - V_{0i})}$$

$$X_3 = \frac{\tau_{si} a s_i L_i}{P_{0i}} - \frac{\tau_{ri} a r_i L_i}{P_{0i}} - \frac{\dot{m}_0 P_{0i}}{P_{0i-1}^2 - P_{0i}^2}$$

$$(V_{0i} - V_{0i-1}) P_{0i} + \frac{\dot{m}_0}{\pi} \mu_{li} (4S_i - 5)$$

$$\left[\frac{1}{P_{0i}} \left(\frac{\gamma - 1}{\gamma} \right) \left(\frac{P_{0i-1}}{P_{0i}} \right)^{\frac{\gamma - 1}{\gamma}} \right] (V_{0i} - V_{0i-1})$$

$$X_4 = \frac{\dot{m}_0 P_{0i-1}}{P_{0i-1}^2 - P_{0i}^2} (V_{0i} - V_{0i-1})$$

$$- \frac{\dot{m}_0}{\pi} (V_{0i} - V_{0i-1}) \mu_{li} (4S_i - 5)$$

$$\left[\frac{1}{P_{0i}} \left(\frac{\gamma - 1}{\gamma} \right) \left(\frac{P_{0i-1}}{P_{0i}} \right)^{-\frac{1}{\gamma}} \right]$$

$$X_5 = \frac{-\dot{m}_0}{Cr_i} (V_{0i} - V_{0i-1}) - \frac{m\tau_{si}as_iL_iD_i}{2(Cr_i + B_i)^2} + \frac{\tau_{ri}armr_iL_iD_i}{2(Cr_i + B_i)^2}$$

A_{i-1} Matrix

$$a_{1,2} = a_{2,1} = a_{3,4} = a_{4,3} = G_4$$

$$a_{5,2} = a_{6,1} = a_{7,4} = a_{8,3} = X_4$$

$$a_{5,6} = a_{6,5} = a_{7,8} = a_{8,7} = -\dot{m}_0$$

The remaining elements are zero.

A_i Matrix

$$a_{1,1} = G_1 \left(\omega + \frac{V_{0i}}{Rs} \right) \quad a_{2,2} = -G_1 \left(\omega + \frac{V_{0i}}{Rs} \right)$$

$$a_{3,3} = G_1 \left(\frac{V_{0i}}{Rs} - \omega \right) \quad a_{4,4} = -G_1 \left(\frac{V_{0i}}{Rs} - \omega \right)$$

$$a_{1,2} = a_{2,1} = a_{3,4} = a_{4,3} = G_3$$

$$a_{5,2} = a_{6,1} = a_{7,4} = a_{8,3} = X_3$$

$$a_{5,1} = a_{7,3} = \frac{A_{0i}}{Rs} \quad a_{6,2} = a_{8,4} = -\frac{A_{0i}}{Rs}$$

$$a_{5,5} = X_1 \left(\omega + \frac{V_{0i}}{Rs} \right) \quad a_{6,6} = -X_1 \left(\omega + \frac{V_{0i}}{Rs} \right)$$

$$a_{7,7} = X_1 \left(\frac{V_{0i}}{Rs} - \omega \right) \quad a_{8,8} = X_1 \left(\omega - \frac{V_{0i}}{Rs} \right)$$

$$a_{5,6} = a_{6,5} = a_{7,8} = a_{8,7} = X_2$$

$$a_{1,5} = a_{3,7} = G_1 \frac{P_{0i}}{Rs}$$

$$a_{2,6} = a_{4,8} = -G_1 \frac{P_{0i}}{Rs}$$

The remaining elements are zero.

A_{i+1} Matrix

$$a_{1,2} = a_{2,1} = a_{3,4} = a_{4,3} = G_5$$

APPENDIX B

Separation of the Continuity and Momentum Equations and Definition of the System Matrix Elements

Continuity

$$\cos(\theta + \omega t): G_{1i}P_{si}^+ \left(\omega + \frac{V_{0i}}{Rs} \right) + G_{1i} \frac{P_{0i}}{Rs} V_{si}^+ + G_{3i}P_{ci}^+ + G_{4i}P_{ci-1}^+ + G_5P_{ci+1}^+ = \frac{G_{6i}}{2} (a-b)$$

$$\sin(\theta + \omega t): -G_{1i}P_{ci}^+ \left(\frac{V_{0i}}{Rs} + \omega \right) - G_{1i} \frac{P_{0i}}{Rs} V_{ci}^+ + G_{3i}P_{si}^+ + G_{4i}P_{si-1}^+ + G_{5i}P_{si+1}^+ = \frac{G_{6i}}{2} \left(\frac{V_{0i}}{Rs} + \omega \right) (b-a)$$

$$\cos(\theta - \omega t): G_{1i} \left(\frac{V_{0i}}{Rs} - \omega \right) P_{si}^- + G_{1i} \frac{P_{0i}}{Rs} V_{si}^- + G_{3i}P_{ci}^- + G_{4i}P_{ci-1}^- + G_{5i}P_{ci+1}^- = \frac{G_{6i}}{2} (a+b)$$

$$\sin(\theta - \omega t): G_{1i} \left(\omega - \frac{V_{0i}}{Rs} \right) P_{ci}^- - \frac{G_{1i}P_{0i}}{Rs} V_{ci}^- + G_{3i}P_{si}^- + G_{4i}P_{si-1}^- + G_{5i}P_{si+1}^- = \frac{G_{6i}}{2} \left(\omega - \frac{V_{0i}}{Rs} \right) (a+b)$$

Momentum

$$\cos(\theta + \omega t): X_{1i} \left(\omega + \frac{V_{0i}}{Rs} \right) V_{si}^+ + \frac{A_{0i}}{Rs} P_{si}^+ + X_{2i}V_{ci}^+ - \dot{m}_0 V_{ci-1}^+ + X_{3i}P_{ci}^+ + X_{4i}P_{ci-1}^+ = \frac{X_{5i}}{2} (b-a)$$

$$\sin(\theta + \omega t): -X_{1i} \left(\omega + \frac{V_{0i}}{Rs} \right) V_{ci}^+ - \frac{A_{0i}}{Rs} P_{ci}^+ + X_{2i}V_{si}^+ - \dot{m}_0 V_{si-1}^+ + X_{3i}P_{si}^+ + X_{4i}P_{si-1}^+ = 0$$

$$\cos(\theta + \omega t): X_{1i} \left(\frac{V_{0i}}{Rs} - \omega \right) V_{si}^- + \frac{A_{0i}}{Rs} P_{si}^- + X_{2i}V_{ci}^- - \dot{m}_0 V_{ci-1}^- + X_{3i}P_{ci}^- + X_{4i}P_{ci-1}^- = \frac{-X_{5i}}{2} (a+b)$$

$$\sin(\theta - \omega t): X_{1i} \left(\omega - \frac{V_{0i}}{Rs} \right) V_{ci}^- - \frac{A_{0i}}{Rs} P_{ci}^- + X_{2i}V_{si}^- - \dot{m}_0 V_{si-1}^- + X_{3i}P_{si}^- + X_{4i}P_{si-1}^- = 0$$

The remaining elements are zero.

B and C column vectors:

$$B = \begin{bmatrix} \frac{G_6}{2} \\ -\frac{G_2}{2} \left(\frac{V_{0i}}{Rs} + \omega \right) \\ \frac{G_6}{2} \\ \frac{G_2}{2} \left(\omega - \frac{V_{0i}}{Rs} \right) \\ \frac{-X_5}{2} \\ 0 \\ \frac{-X_5}{2} \\ 0 \end{bmatrix} \quad C = \begin{bmatrix} \frac{-G_6}{2} \\ \frac{G_2}{2} \left(\frac{V_{0i}}{Rs} + \omega \right) \\ \frac{G_6}{2} \\ \frac{G_2}{2} \left(\omega - \frac{V_{0i}}{Rs} \right) \\ \frac{X_5}{2} \\ 0 \\ \frac{-X_5}{2} \\ 0 \end{bmatrix}$$